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Real-world Evidence for Comparative Safety of Second-Line Antihyperglycemic Agents in Older Adults with Type 2 Diabetes

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ABSTRACT

As prescribing of newer antihyperglycemic agents expands, there remains limited comparative safety data for older adults—a population particularly vulnerable to adverse drug events and underrepresented in clinical trials. We conduct a multinational cohort study using nine databases from the U.S. and Europe. Among adults aged ≥65, we compare eighteen safety outcomes across four classes of second-line antihyperglycemic agents. We use propensity score adjustment, empirical calibration, and prespecified diagnostics and estimated hazard ratios. In a cohort of 1.8 million, both GLP-1 receptor agonists and SGLT2 inhibitors are linked to significantly lower risks of hypoglycemia and hyperkalemia than sulfonylureas. GLP-1 receptor agonists show lower risks of peripheral edema than DPP-4 inhibitors. However, SGLT2 inhibitors are associated with a higher risk of diabetic ketoacidosis compared to GLP-1 receptor agonists and sulfonylureas. These results support informed, safety-conscious prescribing in the older adults underrepresented in clinical trials yet highly susceptible to adverse events.

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INTRODUCTION

As the global population ages, the burden of type 2 diabetes mellitus among older adults continues to rise. According to global estimates, approximately 19.3% of adults aged over 65 years, representing 135.6 million individuals, were living with diabetes in 2019, and this number is projected to increase to 276.2 million by 2045[1]. However, treating this population is challenging, as older adults are uniquely vulnerable to adverse drug events due to multimorbidity, frailty, and polypharmacy.[2,3] Yet, they remain underrepresented in randomized trials. The small sample sizes and selective populations in randomized trials limit their ability to detect rare, severe, or population-specific safety events, leaving clinicians with limited guidance on how to weigh safety considerations when initiating second-line therapy in real-world settings. Furthermore, much of the existing evidence is fragmented or focused on single outcomes, rather than offering a comprehensive assessment of safety across drug classes.[4,5,6]

Clinical guidelines have traditionally recommended a stepwise treatment approach that begins with lifestyle modification and metformin therapy.[7] However, antihyperglycemic agents, including glucagon-like peptide 1 receptor agonists (GLP1RAs), sodium-glucose cotransporter-2 inhibitors (SGLT2Is), dipeptidyl peptidase-4 inhibitors (DPP4Is), and sulfonylureas (SUs), play a more pivotal role in achieving glycemic control than before. These therapies are commonly employed as second-line options for patients who do not reach target glucose levels with first-line treatment, and they are also considered as first-line agents in individuals with comorbidities or clinical indications.[7] In particular, GLP1RAs and SGLT2Is have gained substantial attention for their cardiovascular and renal benefits, prompting guideline endorsements for broader use in high-risk patients.[3,8] However, despite increasing uptake, a critical knowledge gap persists: the comparative safety of these agents, particularly in older adults, remains insufficiently understood.

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The need to address this gap is urgent. As prescribing patterns increasingly shift toward newer agents such as GLP1RA[9] and the older adult population continues to grow, clinicians are often required to make treatment decisions with incomplete safety data for those at highest risk. Without robust comparative evidence, efforts to personalize diabetes care and minimize preventable harm remain limited. Comprehensive, up-to-date evaluations of the safety of second-line antihyperglycemic agents in older adults are therefore essential to inform safer prescribing, reduce avoidable harms, and support evidence-based, patient-centered care in a population frequently underrepresented in clinical trials yet central to everyday clinical practice.

Here, to address this unmet need, we leverage the LEGEND-T2DM initiative[10,11,12], a multinational, large-scale, real-world analytics using a harmonized data model, to evaluate the comparative safety of four major second-line drug classes: GLP1RAs, SGLT2Is, DPP4Is, and SUs. Using rigorous methods including large-scale propensity score adjustment, empirical calibration, and pre-specified diagnostics, we systematically assess a broad range of clinically relevant metabolic, organ system-specific, and systemic safety outcomes.

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RESULTS

Characteristics

Nine international databases comprised five insurance claims databases from the US and four electronic health records (EHRs) from the US, Germany, and Spain[13,14,15,16] were included in this study (Supplementary Table 1). Following pre-specified eligibility criteria (Figure 1), a total of 1,808,003 adults aged 65 years or older with type 2 diabetes who were prescribed a secondary antihyperglycemic agent met the eligibility criteria. There were more male patients with 984,590 (54.5%) than female patients, and 1,211,729 (67.0%) were aged 65-74, and 596,274 (32.9%) were

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aged 75 or older. Among these, 73,603 (4.1%) patients initiated a GLP1RA, 173,465 (9.6%) initiated a SGLT2I, 485,016 (26.8%) initiated a DPP4I, and 1,075,919 (59.5%) received a SU as second-line therapy. The most commonly prescribed agents included dulaglutide (45.4%), semaglutide (36.1%), and exenatide (17.8%) among GLP1RAs; empagliflozin (58.6%), dapagliflozin (22.0%), and canagliflozin (18.9%) among SGLT2Is; sitagliptin (84.3%) among DPP4Is; and glipizide (53.9%) and glimepiride (35.3%) among SUs.

In the on-treatment follow-up, the median days of follow-up periods ranged from 55 to 202 days for GLP1RAs and 59 to 246 days for SGLT2Is, with longer follow-up durations observed for DPP4Is (86–719 days) and SUs (84–741 days) (Table 1). Table 2 presents baseline characteristics for the GLP1RA and SGLT2I groups in the US Open Claims database before and after propensity score stratification. Detailed baseline characteristics and standardized mean differences (SMDs) for all comparisons across nine databases before and after propensity score adjustment are provided in Supplementary Tables 2–36.

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Study Diagnostics

We applied prespecified study diagnostics to ensure reliability, including requirements for empirical equipoise, appropriate covariate balance after propensity score adjustment, and sufficient statistical power based on a minimum number of patients (at least a thousand patients in each group), and a minimum detection relative risk (MDRR). Only treatment comparisons in each database meeting all diagnostic criteria were included in the meta-analyses. Figure 2 shows whether the results from each database passed the diagnostic criteria for the synthesis of hyperkalemia meta-analysis results. For example, for the hyperkalemia outcome, results comparing GLP1RAs and SGLT2Is were generated from four databases, and these four met all

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diagnostic criteria and were ultimately included in the meta-analysis. Similarly, in the GLP1RA vs. DPP4I comparison, the OptumDOD and MDCR databases did not meet the covariate balance criteria and were therefore excluded from the final meta-analysis.

Following study diagnostics, the number of databases included for each drug comparison varied across study outcomes. For GLP1RA comparisons, a median of 2.0 (interquartile range [IQR] 1.0–2.0) databases were included for GLP1RA vs. DPP4I, and 1.0 (1.0–1.0) for GLP1RA vs. SU. For SGLT2I comparisons, a median of 4.0 (IQR 2.0–4.0) databases were included for SGLT2I vs. GLP1RA, 6.0 (5.0–7.0) for SGLT2I vs. DPP4I, and 3.0 (3.0–3.0) for SGLT2I vs. SU. Finally, for the DPP4I vs. SU comparison, a median of 6.5 (5.0–8.0) databases were included. Detailed diagnostic criteria—including thresholds for empirical equipoise, SMD, and MDRR—for each database-comparison pair are available in Supplementary Figure. 36–53.

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Safety Outcomes

For eighteen safety outcomes, meta-analytic results from the primary analysis (on-treatment follow-up with propensity score stratification) are presented in Figure 3. Overall, both GLP1RAs and SGLT2Is were associated with more favorable safety profiles across multiple outcomes compared with SUs and DPP4Is (Supplementary Tables 108–215).

GLP1RA use was associated with a substantially lower risk of hypoglycemia compared to SU, similar to SGLT2I and DPP4I (Hazard ratio [HR] 0.21 [95% CI, 0.16–0.28], HR 0.20 [0.12–0.34], and HR 0.26 [0.21–0.32], respectively). Additionally, both GLP1RAs and SGLT2Is had significantly lower risks of hyperkalemia compared to SU (HR 0.63 [0.50–0.81] and 0.75 [0.63–0.90], respectively), with no meaningful difference observed between the two newer classes. SGLT2Is, in particular, demonstrated lower risk of abnormal weight gain (HR 0.57 [0.45–0.71])

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and higher incidence of abnormal weight loss (HR 1.41 [1.18–1.67]) than SUs, reflecting their known weight-reducing properties. For peripheral edema, both GLP1RAs and SGLT2Is demonstrated lower risks than DPP4Is (HR 0.81 [0.72–0.90] and 0.69 [0.57–0.82], respectively), and were also favorable compared to SUs (HR 0.72 [0.58–0.90] and 0.62 [0.46–0.84], respectively).

However, SGLT2Is showed a significantly increased risk compared to GLP1RAs (HR 2.03 [1.38–2.99]) and SUs (HR 1.64 [1.27–2.11]) for diabetic ketoacidosis. GLP1RAs had shown high risks of nausea (HR 1.58 [1.45–1.72]) and vomiting (HR 1.58 [1.44–1.74]) compared to SGLT2Is. No significant differences across drug classes were observed for bone fracture, joint pain, lower extremity amputation, or venous thromboembolism. Importantly, SGLT2Is were associated with a significantly lower all-cause mortality risk compared to SUs (HR 0.49 [0.35–0.68]), reinforcing the growing evidence supporting their long-term safety benefits in older adults.

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Sensitivity Analysis

Findings from sensitivity analyses using an on-treatment follow-up design with additionally censored patients at treatment escalation are presented in Supplementary Tables 216–323 and were largely consistent with the primary analysis. GLP1RAs continued to show a favorable safety profile, including a significantly lower risk of hyperkalemia compared to SUs (HR 0.62 [0.49–0.77]) and reduced risk of peripheral edema compared to SUs (HR 0.71 [0.58–0.86]). SGLT2Is also maintained a lower risk of peripheral edema compared to both DPP4Is and SUs (HR 0.70 [0.60–0.83] and HR 0.65 [0.52–0.80], respectively).

However, SGLT2Is also remained associated with a significantly increased risk of diabetic ketoacidosis relative to both GLP1RAs and SUs (HR 1.99 [1.31–3.01] and HR 1.70

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[1.28–2.27], respectively). In contrast, GLP1RAs continued to demonstrate inferior gastrointestinal tolerability, with significantly higher risks of nausea (HR 1.63 [1.50–1.77]) and vomiting (HR 1.62 [1.47–1.79]) compared to SGLT2Is. Across all drug classes, the risk of hypoglycemia remained significantly lower compared to SU, further reinforcing the advantages of GLP1RAs (HR 0.21 [0.16–0.27]), SGLT2Is (HR 0.17 [0.09–0.30]), and DPP4Is (HR 0.17 [0.13–0.21]).

In the intention-to-treat analysis, which introduces greater bias toward the null due to persistent exposure misclassification, fewer relative risk estimates met the Bonferroni-corrected significance threshold. Nonetheless, key safety signals remained consistent (Figure 4 and Supplementary Tables 108–215). GLP1RAs continued to show higher risks of nausea (HR 1.24 [1.10–1.40]) and vomiting (HR 1.22 [1.10–1.34]) compared to SGLT2Is. Importantly, GLP1RAs and SGLT2Is showed a significantly lower risk of all-cause mortality compared to DPP4Is (HR 0.56 [0.39–0.79] and 0.79 [0.67–0.93], respectively), highlighting their potential long-term benefit in older adults.

Results of the meta-analyses, pooled separately by database type (Claims vs. EHR) and region (US vs. non-US), are presented in Supplemental Figure. 58–59. Although the meta-analysis results from Claims and EHR databases showed a high degree of agreement in their significance, the results between US and non-US databases were generally consistent only for the DPP4I vs. SU comparison due to the limited numbers of GLP1RA and SGLT2I cohorts in the non-US database. In a leave-one-database-out meta-analysis that exclude one database for each outcome conducted to determine the influence of individual databases, most comparisons showed consistent significances (Supplemental Figure. 60–95).

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DISCUSSION

This study provides new and comprehensive insight into the comparative safety of second-line antihyperglycemic agents in older adults—a population at high risk for drug-related complications yet historically underrepresented in clinical trials. By analyzing over 1.8 million patients across multiple international databases, we found that GLP1RAs and SGLT2Is generally offer more favorable safety profiles compared to SUs and DPP4Is. Importantly, we identified clinically relevant differences across a broad set of safety outcomes—not just the well-known risks like hypoglycemia or gastrointestinal side effects,[17] but also less frequently assessed events in clinical trials such as peripheral edema, diabetic ketoacidosis, and hyperkalemia.[18,19,20,21,22] These findings help move the field beyond efficacy comparisons to a more nuanced, safety-centered understanding of treatment choices in older adults with type 2 diabetes.

The observed differences in safety profiles across drug classes likely reflect their distinct pharmacologic mechanisms and how those interact with age-related vulnerabilities. Sulfonylureas, which stimulate insulin secretion irrespective of glucose levels, predictably increase the risk of hypoglycemia, especially in older adults with fluctuating food intake or renal impairment[23]. The weight gain and edema associated with sulfonylureas may be tied to fluid retention and insulin-mediated anabolic effects[24,25]. In contrast, SGLT2Is promote glucosuria and osmotic diuresis, which can reduce body weight and blood pressure but also predispose to volume depletion in frail or volume-sensitive patients.[26] The lower incidence of peripheral edema observed with GLP1RAs and SGLT2Is may be due to their favorable renal and vascular effects[27,28]. In our findings, SGLT2I was associated with an elevated risk of diabetic ketoacidosis, which previous retrospective studies have similarly identified[29,30,31,32,33] but rarely detected in RCTs.[18,19] In a context where previous studies have limited evidence on hyperkalemia with SGLT2I and GLP1RA,[20,21,22] our study

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showed a significantly lower risk of hyperkalemia with SGLT2I and GLP1RA. However, potential differences between SGLT2I and GLP1RA suggested in previous literature were not observed in this study.[21] Bone fractures are an outcome of particular relevance in older adults, and a previous trial suggested that SGLT2I may increase this risk. In contrast, we did not observe any significant differences across comparison pairs, which is consistent with findings from prior observation studies[32,34]. GLP1RAs, acting through central and gastrointestinal pathways, are known to cause nausea and vomiting, especially during initiation[35]. Incretin-based glucose lowering agents have been reported to be potentially associated with acute pancreatitis[36,37,38], however, in our study, we observed no significant difference in the risk of acute pancreatitis between SGLT2I and GLP1RA. Notably, several prior studies have suggested that liraglutide may carry a stronger association with pancreatitis compared other GLP1RAs[39,40,41,42]. Because the proportion of liraglutide use in our GLP1RA was very low, individual ingredient-specific analyses will be needed to more definitively evaluate these potential differences. Finally, the signals for reduced all-cause mortality among GLP1RA and SGLT2I users align with emerging evidence of their cardioprotective and anti-inflammatory properties, which may translate into broader safety benefits in older populations[43,44,45]. These results emphasize that prioritizing SGLT2Is and GLP1RAs as second-line treatment in older adults over the current pattern of prescribing DPP4s and SUs with type 2 diabetes may be potentially beneficial[9].

Our study extends the literature in three important ways. First, we focus explicitly on safety outcomes in older adults, a population frequently excluded or underrepresented in randomized controlled trials but disproportionately vulnerable to adverse drug events due to multimorbidity, frailty, and polypharmacy. Prior studies have either aggregated findings across age groups or relied on post hoc subgroup analyses, limiting the ability to draw robust conclusions specific to older

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adults. Second, we conduct a comprehensive, class-wide comparison of four major second-line antihyperglycemic drug classes rather than focusing on isolated pairwise comparisons or single adverse events. Previous real-world studies have typically concentrated on one or two outcomes (e.g., hypoglycemia or heart failure) or compared only a subset of these drug classes. Our study fills this gap by systematically examining 18 clinically relevant outcomes, allowing for a more balanced and practical evaluation of the relative safety of each class. Third, we applied a rigorous and transparent analytic framework, using harmonized multinational real-world data, large-scale propensity score adjustment, and empirical calibration. While prior observational studies often rely on a single database or apply variable methods across studies, our approach, anchored in the LEGEND and OHDSI frameworks, ensures consistency, reproducibility, and generalizability of the findings across diverse healthcare settings. This methodological strength not only confirms previously documented risks (such as hypoglycemia with sulfonylureas) but also uncovers less-recognized differences, such as the gastrointestinal tolerability advantages of SGLT2Is over GLP1RAs and the mortality benefits of SGLT2Is compared to sulfonylureas. These nuanced insights are particularly valuable for tailoring treatment decisions in older adults, where safety considerations often take precedence over modest differences in glycemic control.

These findings have important clinical and policy implications. First, they reinforce current guideline recommendations, such as those from the American Diabetes Association and European Association for the Study of Diabetes that prioritize GLP1RAs and SGLT2Is over SUs and DPP4Is, particularly in older adults.[46,47] These preferences have largely been based on cardiometabolic benefits, but our results add further justification by demonstrating superior safety profiles across a wide range of adverse outcomes. In clinical practice, this means that clinicians should prioritize newer agents—especially in patients with multiple comorbidities, frailty, or a history of adverse

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drug reactions. Second, the findings highlight the need to reevaluate entrenched prescribing patterns that continue to favor SUs, which pose disproportionate risks of hypoglycemia, weight gain, and mortality in older adults despite their low cost. These patterns may reflect legacy habits, formulary limitations, or clinician concerns about the cost or side effects of newer agents. Addressing these barriers may require targeted educational efforts, updated formularies, or value-based pricing strategies to ensure that safety and effectiveness guide treatment choices. Third, these results can inform shared decision-making by providing patients and caregivers with more nuanced information about potential risks and tolerability profiles. For example, the lower rates of nausea and vomiting associated with SGLT2Is compared to GLP1RAs may be meaningful to patients who previously discontinued medications due to gastrointestinal side effects. Likewise, awareness of the increased risk of diabetic ketoacidosis associated with SGLT2Is may prompt closer monitoring or alternative choices in certain subgroups, such as those with low insulin reserve, volume depletion, or increased susceptibility to infection. At a systems level, this study underscores the feasibility and value of multinational, large-scale real-world safety surveillance. The use of harmonized data and transparent, reproducible methods provides a model for generating actionable evidence that is both rigorous and patient-centered. These approaches can be extended to other drug classes, populations, or outcomes, supporting a more responsive and evidence-driven healthcare system, especially for populations such as older adults who are often excluded from clinical trials.

Several limitations should be considered when interpreting our findings. First, as with all observational studies, residual confounding cannot be fully excluded, despite our use of large-scale propensity score adjustment and empirical calibration. Second, although this study applied rigorous study diagnostics to ensure the internal validity and credible estimation, these procedures

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may introduce limitations in generalizability. For example, only U.S. databases met all diagnostics for the GLP1RA comparisons; the corresponding results may not extend to older adults in Europe and Asia. In addition, focusing exclusively on incident outcomes, individuals with a prior history of the outcome were excluded; therefore, our estimate should not be interpreted as reflecting risk of recurrence among patients with pre-existing outcome. Third, some key information was not included due to limitations in the data sources. For example, drug exposure was based on prescription rather than dispensing records and may not reflect actual patients' compliance. Furthermore, the lack of laboratory data of patients-reported outcomes, such as a glycated glucose or electrolyte levels, limited of the detail of some safety assessments. Fourth, we were unable to address the emerging questions after the protocol was developed. Certain outcomes, such as cognitive decline, neuropsychiatric outcomes, and ophthalmological complications, were not included. Furthermore, the anticipated small sample size precluded conducting interesting subgroup analyses (e.g., age ≥75 years or by sex) in the older adult population. While these limitations may attenuate some findings or leave specific questions unanswered, they do not diminish the overall value of this large-scale, real-world evaluation of safety in a high-risk population.

In conclusion, GLP1RAs and SGLT2Is are generally safer than SUs and DPP4Is in older adults with type 2 diabetes, specifically with lower risks of hypoglycemia, hyperkalemia, edema, and mortality. These findings provide robust, real-world evidence to support safer prescribing choices in a high-risk population. Incorporating these safety considerations into clinical decision-making can improve outcomes and reduce avoidable harm in older adults.

METHODS

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Ethical considerations

All informed patient consent was waived or deemed not applicable due to the retrospective, de-identified nature of the data. Ethical approvals and exemptions were obtained as required for each data source. For CCAE, MDCR, MDCD, OptumDOD, and OptumEHR databases, the New England Independent Review Board determined the study to be exempt from ongoing IRB approval. Because the data are fully de-identified, the project does not constitute human subject research, which inherently waives the need for informed consent. For US Open Claims and IQVIA Disease Analyzer, Germany (DAG) databases, no ethics committee oversight is applicable. These are commercially available, de-identified data assets; their usage is governed by compliance with Data Use Agreements rather than institutional ethics review.

For the VA-OMOP database, the study was overseen by the VA Salk Lake City Health Care System Research and Development Committee. The University of Utah IRB (IRB #00192565) determined the study met the criteria for exemption from full ethics review under Exemption Category 4 (45 CRF 46.104(d)) and officially approved the Waiver of HIPAA Authorization (45 CRF 164.512(i)), which serves as the waiver for informed consent.

Lastly, for the SIDIAP database, the study was approved by the Clinical Research Ethics Committee (CElm) of the IDIAP Jordi Gol (Project code: 22/020-P). The committee certified compliance with the Declaration of Helsinki and applicable data protection regulations, thereby waiving the requirement for individual informed consent for this retrospective anonymized study.

Data source and study design

This study is part of the LEGEND-T2DM initiative, led by the Observational Health Data Sciences and Informatics (OHDSI) research collaborative.[48,49] We conducted a retrospective observational

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cohort study using international databases standardized to the Common Data Model (CDM).[13] All databases mapped to version 5.3.1 of the Observational Medical Outcomes Partnership (OMOP) CDM.[50] Basic information, including data coverage periods and contributing entities for each source, is presented in Supplementary Table 1. Potential overlap of patients across de-identified U.S. data sources could not be evaluated or removed due to the fully de-identified nature of the data.

Detailed definitions of the study population, exposures, outcomes, follow-up strategies, and statistical methods were prespecified, registered, and published prior to study execution.[48,49] The study utilized OHDSI's Health Analytics Data-to-Evidence Suite, a suite of open-source tools for epidemiologic research[51]. All study results were fully disclosed to support transparency and open science. The study followed the LEGEND principles and was reported in accordance with STROBE guidelines[11,52].

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Study population and exposure

The study population included older adults (aged ≥65 years) with a diagnosis of type 2 diabetes who initiated a second-line antihyperglycemic agent following first-line metformin monotherapy (Figure 4). Second-line agents were defined as GLP1RAs, SGLT2Is, DPP4Is, and SUs.

We employed a new-user, active-comparator design, with the index date defined as the date of the first prescription for a second-line agent. Patients were excluded if they met any of the following criteria: (1) less than one year of continuous observation prior to the index date; (2) a diagnosis of type 1 diabetes or secondary diabetes mellitus; (3) prior use of antihyperglycemic agents other than the specified second-line agents; or (4) fewer than 90 days of metformin monotherapy or evidence of long-term insulin use (defined as ≥30 days of continuous use).

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Study Outcome

All safety outcomes were selected based on the 2018 American Diabetes Association guidelines and prior randomized controlled trials[53]. We evaluated eighteen outcomes, grouped into three categories: (1) Metabolic and endocrine complications (abnormal weight gain, abnormal weight loss, diabetic ketoacidosis, hyperkalemia, hypoglycemia, and hypotension); (2) Organ system complications (acute pancreatitis, nausea, vomiting, diarrhea, bone fracture, joint pain, lower extremity amputation, peripheral edema, genitourinary infection, and photosensitivity); and (3) Systemic complications (all-cause mortality and venous thromboembolism). Outcome definitions have been previously implemented and validated[54]. Detailed definitions of study outcome were presented in the study protocol[48]. For each outcome, individuals with a prior history of the event were excluded.

Follow-up

Follow-up was assessed using three approaches: (1) on-treatment regardless of treatment escalation, (2) on-treatment with additional censoring upon treatment escalation, and (3) intention-to-treat. In the on-treatment approach, patients were followed from the day after the index date until the earliest occurrence of treatment discontinuation, outcome event, or end of observation. Treatment was considered continuous if a new prescription was recorded within 30 days of the previous one. Discontinuation was defined as the absence of a new prescription following the end of the prior prescription. The on-treatment with additional censoring upon treatment escalation approach followed the same rules as on-treatment, but additionally censored follow-up upon initiation of any new antihyperglycemic agent. In contrast, the intention-to-treat approach followed

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patients from the index date until the end of observation or outcome occurrence, regardless of treatment discontinuation or escalation.

Statistical analysis

Large-scale propensity scores were estimated using L1-penalized logistic regression for each treatment comparison within each database, incorporating all available covariates across clinical data domains (including demographics, conditions, drugs, procedures, measurements, devices, and observations) over multiple time windows prior to treatment initiation (index date, 6 months prior, 1 year prior, and any time prior)[55]. Sex was based on coded information from individual data sources. The measurements were included as only the measurement status as a covariate, not the value. Values that did not exist in the database were treated as unobserved, and no separate method (e.g., imputation) was used to supplement missing data.

Confounding was addressed using propensity score stratification into 10 strata and variable-ratio matching for each pairwise comparison between second-line agents. To enhance generalizability, we prioritized reporting results based on propensity score stratification. HR for each outcome were estimated using Cox proportional hazards models within each database. Empirical calibration of HRs was conducted using negative control outcomes to correct for residual systematic error and to maintain nominal statistical properties despite the observational study design[56].

In total, we conducted 38,232 analyses, based on 9 databases × 6 comparison pairs × 118 outcomes (18 safety outcomes plus negative controls) × 2 PS methods × 3 follow-up approaches. Statistical significance was defined as a two-sided p-value <0.05. In addition, we applied Bonferroni correction for multiple testing in the meta-analytic results (Threshold with p <

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0.00277778 from 18 outcomes hypotheses for each comparison), and both unadjusted and adjusted values are reported.

Prior to evidence synthesis, we applied prespecified study diagnostics to ensure reliability. First, to confirm empirical equipoise, at least 25% of patients in each comparison within each database were required to have preference scores between 0.3 and 0.7[57]. Second, all covariates were required to have a maximum SMD <0.15 following PS adjustment, indicating adequate balance. Third, we required a MDRR <4.0 and a minimum sample size of ≥1,000 patients per treatment group to ensure sufficient statistical power[58]. Only comparisons and databases that met all diagnostic criteria were included in the meta-analyses.

Pooled HR estimates were obtained using a random-effects meta-analysis of the empirically calibrated database-specific HRs. All analyses were performed in a controlled computing environment using OHDSI and other R packages (version 4.2.3)[51]. The full analytic code and results are publicly available at: https://github.com/ohdsi-studies/LegendT2dm, https://data.ohdsi.org/LegendT2dmClassEvidenceExplorer/.

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Data Availability

The patient-level raw data used in this study cannot be publicly shared due to strict data privacy regulations, ethical considerations, and commercial Data Use Agreements. The underlying databases (e.g., Merative MarketScan, Optum, IQVIA, VA, SIDIAP) are subject to licensing restrictions and institutional governance. Researchers interested in accessing the raw patient-level data must apply directly to the respective data custodians for access or licensing.

However, to ensure full transparency and enable the verification of our findings, the minimum dataset necessary to interpret and reproduce the research reported in this article is openly available. Comprehensive aggregated summary statistics, baseline characteristics, propensity score diagnostics, and full comparative effectiveness analysis results are transparently published through interactive online repositories. Cohort characteristics analysis results can be accessed at https://data.ohdsi.org/LegendT2dmClassCohortExplorer/ and comparative effectiveness analysis results at https://data.ohdsi.org/LegendT2dmClassEvidenceExplorer/.

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Code Availability

The research protocol and code used in the study were specified before conducting the analysis. The protocol was published on a public website and in a peer-reviewed journal[48,49], and all analysis code is available publicly through a GitHub repository (https://github.com/ohdsi-studies/legendt2dm).

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Author Contributions Statement

RK, MJS, YL, AO, PBR, GH, HMK, and MAS designed the study. CK, FB, CB, TDS, BV, TF, AP, JL, CY, MVZ, and MAS performed statistical analyses. All authors contributed to the interpretation of the results. RK, GH, AN, EM, JYJ, MS, SS, DRM, HMK, MAS, and YL provided significant guidance. CK, YL drafted the initial manuscript. All authors read, contributed significantly, and approved the final manuscript.

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Competing Interests Statement

Dr. Sawano has received partial research funding through Yale University from PolyBio, Pfizer, and Norvatis. He has also received lecture honoraria from Boehringer Ingelheim and Abbott and receives compensation for his role as Execute Associate Editor for JACC, all outside the scope of this work. Dr. Nishimura received support from the National Institute of General Medical Sciences of the National Institute of Health. Dr. Khera is an associate editor at JAMA; and has received support from the National Institutes of Health (awards R01 AG089981, R01 HL167858

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and K23 HL153775) and the Doris Duke Charitable Foundation (award 2022060); has received research support, through Yale University, from Bristol Myers Squibb, Novo Nordisk, and BridgeBio; is a coinventor of U.S. provisional patent applications WO2023230345A1, US20220336048A1, 63/177,117, 63/346,610, 63/428,569, 63/484,426, 63/508,315, 63/580,137, 63/606,203, 63/619,241, and 63/562,335, unrelated to the present work; and is a cofounder of Evidence2Health and Ensight-AI, precision health platforms to improve evidence-based cardiovascular care and cardiovascular diagnostics. Dr. Thangaraj is a coinventor on provisional patent 63/606,203, unrelated to the present work, and is funded by grant 5T32 HL155000-03. Ms. Blacketer, Dr. Ostropolets, Dr. Schuemie, and Dr. Ryan are employees of Johnson & Johnson and shareholders of Johnson & Johnson. Ms. Li and Ms. Yin are employees of IQVIA. Ms. Zandt is an employee of IQVIA and shareholders of IQVIA. Dr. Lau received research fundings from Diabetes UK outside the scope of this work. Dr. Minty discloses a consulting relationship with Orpyx Medical Techonologies, equity ownership in Orpyx Medical Technologies, and status as co-inventor on 2 patent applications (US 20250000440, US 20250000440), all of which are outside the scope of this work. Dr. You reports being a chief executive officer of PHI Digital Healthcare, and grants from Daiichi Sankyo. He is a coinventor of granted Korea Patent DP-2023-1223 and DP-2023-0920, and pending Patent Applications DP-2024-0909, DP-2024-0908, DP-2022-1658, DP-2022-1478, and DP-2022-1365, unrelated to current work. Dr. Man has received support from the C.W. Maplethorpe Fellowship, the National Institute of Health Research, European Commission Framework Horizon 2020, the Hong Kong Research Grant Council, and the Innovation and Technology Commission of the Hong Kong Special Administration Region Government outside the submitted work). Dr. Hripcsak received a grant from Johnson & Johnson for methods work unrelated to this study. Dr. Krumholz has

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All other authors have reported that they have no relationships relevant to the contents of this paper to disclose.

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Table 1. Population size and follow-up time for each drug class within each data source.

Drug class, Total and by database	Patients, N	On-treatment time (days), Median (IQR)	Total follow-up time (days), Median (IQR)
GLP1RA, total	73,603		
MDCR	2135	108 (36-300)	513 (159-1,129)
ODOD	5627	95 (40-219)	345 (124-728)
OEHR	3573	55 (29-102)	527 (273-983)
USOC	62268	105 (50-258)	585 (241-1,130)
SGLT2I, total	173,465		
DAG	2,727	157 (97-372)	598 (293-1,155)
MDCR	3,511	121 (38-344)	419 (156-880)
ODOD	11,619	114 (55-273)	390 (146-799)
OEHR	7,828	59 (29-99)	596 (295-1,081)
SIDIAP	3,746	246 (84-617)	498 (155-875)
USOC	135,495	125 (59-309)	681 (293-1,342)
VA	8,539	144 (71-306)	283 (124-592)
DPP4I, total	485,016		
CCAE	2,089	86 (29-144)	110 (56-180)
DAG	12,158	219 (97-509)	1,174 (564-2,065)
MDCD	1,210	130 (56-330)	834 (357-1,539)
MDCR	17,961	191 (87-491)	815 (333-1,533)
ODOD	27,244	157 (63-375)	832 (342-1,581)
OEHR	23,345	90 (89-194)	1,076 (537-1,827)
SIDIAP	26,465	719 (200-1,464)	1,207 (567-2,137)
USOC	358,700	158 (59-419)	1,575 (806-2,430)
VA	15,844	235 (90-523)	756 (348-1,337)
SU, total	1,075,919		
CCAE	3,916	84 (29-146)	113 (57-189)
DAG	4,149	211 (119-467)	2,043 (1,003-3,325)
MDCD	2,488	126 (48-311)	1,020 (464-1,874)
MDCR	43,297	186 (72-496)	890 (354-1,780)
ODOD	66,309	193 (89-504)	882 (352-1,709)
OEHR	58,387	102 (89-231)	1,219 (587-2,042)
SIDIAP	8,667	741 (188-1,615)	2,879 (1674-3,808)
USOC	741,462	203 (89-547)	1,498 (717-2,419)
VA	147,244	224 (90-544)	2,071 (1,043-3,330)

This table reports the total patient counts and median and interquartile (IQR) times. When executing comparative studies, we included only within-database populations with > 1,000 new users per treatment-arm. Abbreviations: IQR: interquartile range; GLP1RA: glucagon-like peptide-1 receptor agonist; SGLT2I: sodium-glucose cotransporter 2 inhibitor; DPP4I: dipeptidyl peptidase 4 inhibitor; SU: sulfonylurea; CCAE: Merative MarketScan Commercial Claims and Encounter (CCAE); MDCR: Merative MarketScan Medicare Supplemental Database; MDCD: Merative MarketScan Multi-state Medicaid Database; USOC: IQVIA Open Claims; ODOD: Optum Clinformatics Data Mart; DAG: IQVIA Disease Analyzer Germany; OEHR: Optum Electronic Health Records; SIDIAP: Information System for Research in Primary Care; VA: US Department of Venterans Affairs.

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Table 2. Baseline characteristics for SGLT2I (T) and GLP1RA (C) new-users in the USOC data source.

Characteristics	Before adjustment			After matching			After stratification		
	T (%)	C(%)	SMD	T (%)	C(%)	SMD	T (%)	C(%)	SMD
Gender: Female	44.7	57.3	0.25	55.3	54.7	-0.01	48.8	48.3	-0.01
Age group, years									
65-69	43.1	49.1	0.12	44.9	44.9	0.00	44.9	44.9	0.00
70-74	29.2	29.8	0.01	30.1	29.9	0.00	29.5	29.4	0.00
75-79	17.5	14.2	-0.09	14.7	14.8	0.00	16.4	16.6	0.00
80-84	9.5	6.4	-0.12	6.7	6.8	0.00	8.5	8.6	0.00
85-89	0.8	0.5	-0.03	0.5	0.5	0.00	0.7	0.6	-0.01
Medical history: General									
Chronic liver disease	0.9	0.9	-0.01	0.9	0.9	0.00	0.9	0.9	0.00
Chronic obstructive lung disease	7.2	7.1	0.00	7.1	6.9	-0.01	7.3	7.1	0.00
Dementia	1.0	0.8	-0.02	0.9	0.9	0.00	1.0	1.0	0.00
Depressive disorder	6.0	8.8	0.10	8.0	7.7	-0.01	6.9	7.0	0.00
Hyperlipidemia	48.5	46.1	-0.05	45.8	45.6	0.00	47.8	47.8	0.00
Hypertensive disorder	54.8	53.5	-0.03	53.1	52.7	-0.01	54.5	54.6	0.00
Obesity	9.6	16.2	0.20	13.8	13.4	-0.01	11.6	11.9	0.01
Osteoarthritis	18.0	22.1	0.10	20.9	20.5	-0.01	19.3	19.4	0.00
Renal impairment	10.4	9.4	-0.03	9.7	9.2	-0.02	10.2	10.0	-0.01
Rheumatoid arthritis	1.2	1.5	0.03	1.4	1.3	-0.01	1.3	1.3	0.00
Medical history: Cardiovascular disease									
Cerebrovascular disease	4.7	3.9	-0.04	4.0	3.9	0.00	4.5	4.5	0.00
Coronary arteriosclerosis	16.4	11.4	-0.14	12.0	11.7	-0.01	15.0	14.7	-0.01
Heart failure	8.6	5.0	-0.14	5.2	5.0	-0.01	7.6	6.8	-0.03
Ischemic heart disease	7.2	4.6	-0.11	4.8	4.7	-0.01	6.4	6.1	-0.01
Peripheral vascular disease	7.5	6.7	-0.03	6.8	6.6	-0.01	7.3	7.3	0.00
Medical history: Neoplasms									
Malignant neoplastic disease	9.1	8.7	-0.02	8.8	8.5	-0.01	9.0	8.8	-0.01
Malignant tumor of breast	1.5	1.8	0.02	1.8	1.7	0.00	1.6	1.5	-0.01
Malignant tumor of colon	0.4	0.3	-0.01	0.4	0.3	0.00	0.4	0.3	0.00
Malignant tumor of lung	0.3	0.3	-0.01	0.3	0.3	0.00	0.3	0.3	0.00
Malignant tumor of urinary bladder	0.4	0.4	0.00	0.4	0.4	0.00	0.4	0.5	0.01
Primary malignant neoplasm of prostate	2.3	1.8	-0.03	1.9	1.9	0.00	2.2	2.2	0.00
Medication use									
Agents acting on the renin-angiotensin system	74.4	70.2	-0.10	71.0	70.9	0.00	73.2	73.0	0.00
Antibacterials for systemic use	55.4	59.1	0.07	58.5	57.8	-0.01	56.7	56.3	-0.01
Antiinflammatory and antirheumatic products	33.6	36.0	0.05	35.5	34.9	-0.01	34.5	34.0	-0.01
Antithrombotic agents	28.7	22.2	-0.15	22.6	22.7	0.00	26.6	26.4	0.00
Beta blocking agents	47.5	41.1	-0.13	41.8	41.7	0.00	45.5	45.4	0.00
Calcium channel blockers	33.2	31.1	-0.04	31.5	31.5	0.00	32.5	32.8	0.01
Diuretics	47.9	49.4	0.03	49.3	48.5	-0.02	48.6	48.1	-0.01
Drugs for obstructive airway disease	39.5	44.3	0.10	43.4	42.7	-0.01	41.2	41.0	0.00
Lipid modifying agents	81.1	77.0	-0.10	77.7	77.8	0.00	79.8	79.8	0.00
Opioids	22.8	26.2	0.08	25.4	24.8	-0.01	24.0	23.8	-0.01

This table reports the proportion of initiators satisfying selected baseline characteristics and the standardized difference of population proportions before and after propensity score adjustment. Less extreme SMD (<0.15) through matching and stratification suggest improved balance between patient cohorts through adjustment. Abbreviations: SGLT2I: sodium-glucose transporter-2 inhibitor; GLP1RA: glucagon-like peptide-1 receptor agonist; SMD: standardized mean difference.

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FIGURE LEGENDS

Figure 1. Study design that depicts target-comparator pairs, eligibility criteria, and follow-up strategies.
T2DM: type 2 diabetes mellitus; Rx: prescription.

Figure 2. Pre-specified study diagnostics results by target-comparator pairs and database used for hyperkalemia outcome.

This plot shows whether cohort diagnostic criteria are met before and after propensity score adjustment for each database. Only exposure-comparator pairs with more than 1,000 individuals in each group within the database are displayed. The Equipoise represents the percentage of the preference score in the corresponding interval (0.3–0.7) for each database. A database passes diagnostics if it exceeds 25%. The Max SMD represents the maximum SMD value after propensity score adjustment, and a database passes diagnostics if it is less than 0.15. An MDRR of 4 or less passes diagnostics.

GLP1RA: glucagon-like peptide-1 receptor agonist; SGLT2I: sodium-glucose transporter-2 inhibitor; DPP4I: dipeptidyl peptidase 4 inhibitor; SU: sulfonylurea; CCAE: Merative MarketScan Commercial Claims and Encounter (CCAE); MDCR: Merative MarketScan Medicare Supplemental Database; MDCD: Merative MarketScan Multi-state Medicaid Database; USOC: IQVIA Open Claims; ODOD: Optum Clinformatics Data Mart; DAG: IQVIA Disease Analyzer Germany; OEHR: Optum Electronic Health Records; SIDIAP: Information System for Research in Primary Care; VA: US Department of Venterans Affairs; SMD: standardized mean difference; MDRR: Minimum detectable relative risk.

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Figure 3. Meta-analytic safety profiles comparing new users of SGLT2I to GLP1RA, DPP4I, and SU across 18 outcomes.

Points and lines identify calibrated HR estimates from meta-analyses with their 95% CIs, respectively. All tests for the meta-analysis were two-sided. The result for all-cause mortality in the GLP1RA vs SU comparison was not presented because there was no valid result after the study diagnostic process. The precise values for point estimates and confidence intervals, as well as the number of databases included in the meta-analysis, were provided in Supplementary Tables 108–215, respectively.

PS: propensity score; GLP1RA: glucagon-like peptide-1 receptor agonist; SGLT2I: sodium-glucose transporter-2 inhibitor; DPP4I: dipeptidyl peptidase 4 inhibitor; SU: sulfonylurea.

Figure 4. Safety profiles comparing new users of SGLT2I to GLP1RA, DPP4I, and SU across 18 outcomes from the intent-to-treat follow-up.

Points and lines identify calibrated HR estimates from meta-analyses with their 95% CIs, respectively. All tests for the meta-analysis were two-sided. The result for all-cause mortality in the GLP1RA vs SU comparison was not presented because there was no valid result after the study diagnostic process. The precise values for point estimates and confidence intervals, as well as the number of databases included in the meta-analysis, were provided in Supplementary Tables 108–215, respectively.

PS: propensity score; GLP1RA: glucagon-like peptide-1 receptor agonist; SGLT2I: sodium-glucose transporter-2 inhibitor; DPP4I: dipeptidyl peptidase 4 inhibitor; SU: sulfonylurea.

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Editor's Summary

Here the authors report real-world evidence through a retrospective analysis of a multinational cohort of 1.8M older adults showing that GLP1RAs and SGLT2 inhibitors carry lower risk for hyperkalemia than sulfonylureas. However, SGLT2 inhibitors increased risk of ketoacidosis. Findings support safety-conscious prescribing for older adults, who are often underrepresented in clinical trials.

Peer Review Information: *Nature Communications* thanks the anonymous reviewer(s) for their contribution to the peer review of this work. A peer review file is available.

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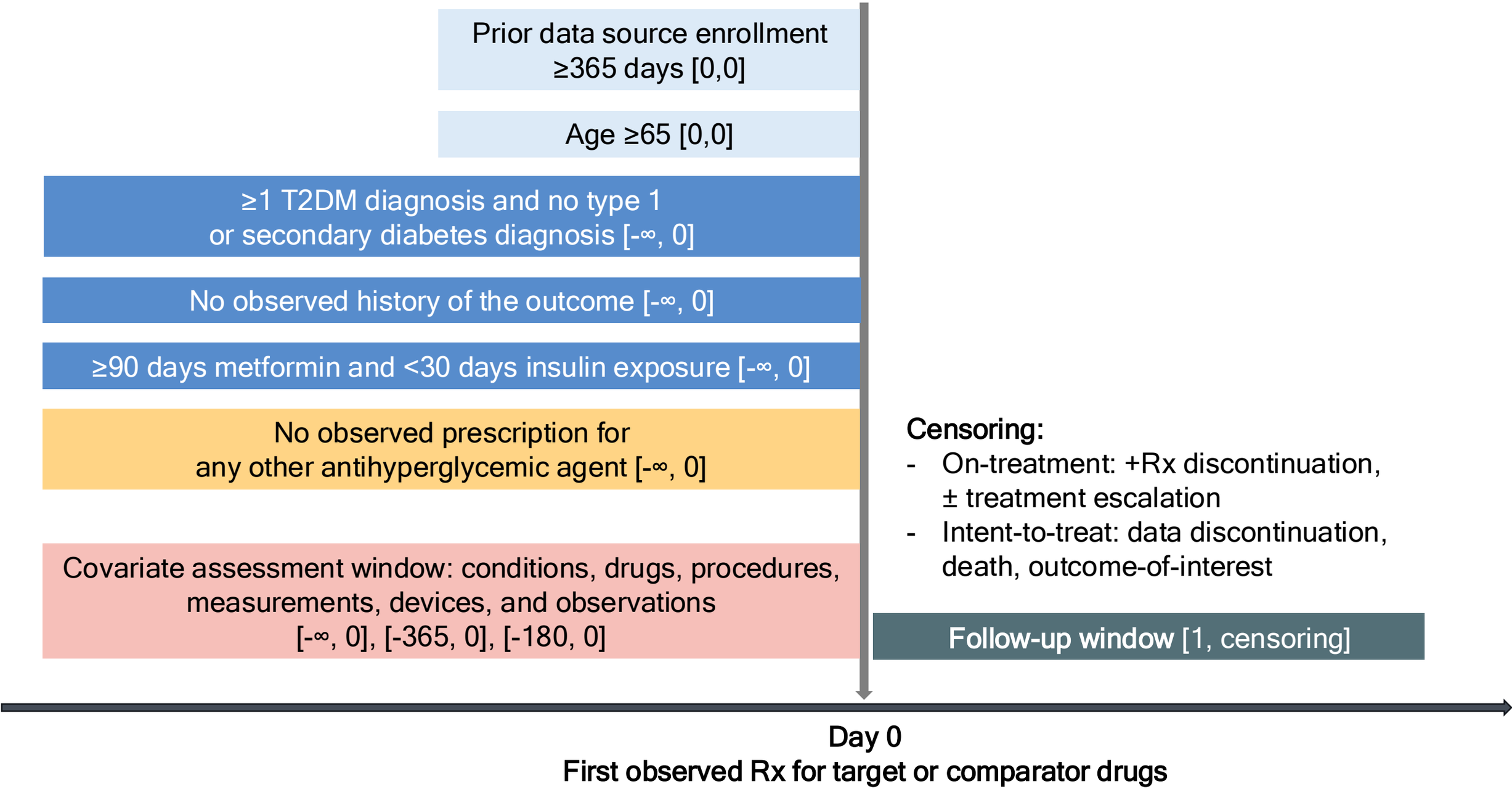
Prior data source enrollment
≥365 days [0,0]
Age ≥65 [0,0]
≥1 T2DM diagnosis and no type 1
or secondary diabetes diagnosis [-∞, 0]
No observed history of the outcome [-∞, 0]
≥90 days metformin and <30 days insulin exposure [-∞, 0]
No observed prescription for
any other antihyperglycemic agent [-∞, 0]
Covariate assessment window: conditions, drugs, procedures,
measurements, devices, and observations
[-∞, 0], [-365, 0], [-180, 0]
Censoring:
- On-treatment: +Rx discontinuation,
± treatment escalation
- Intent-to-treat: data discontinuation,
death, outcome-of-interest
Follow-up window [1, censoring]
Day 0
First observed Rx for target or comparator drugs

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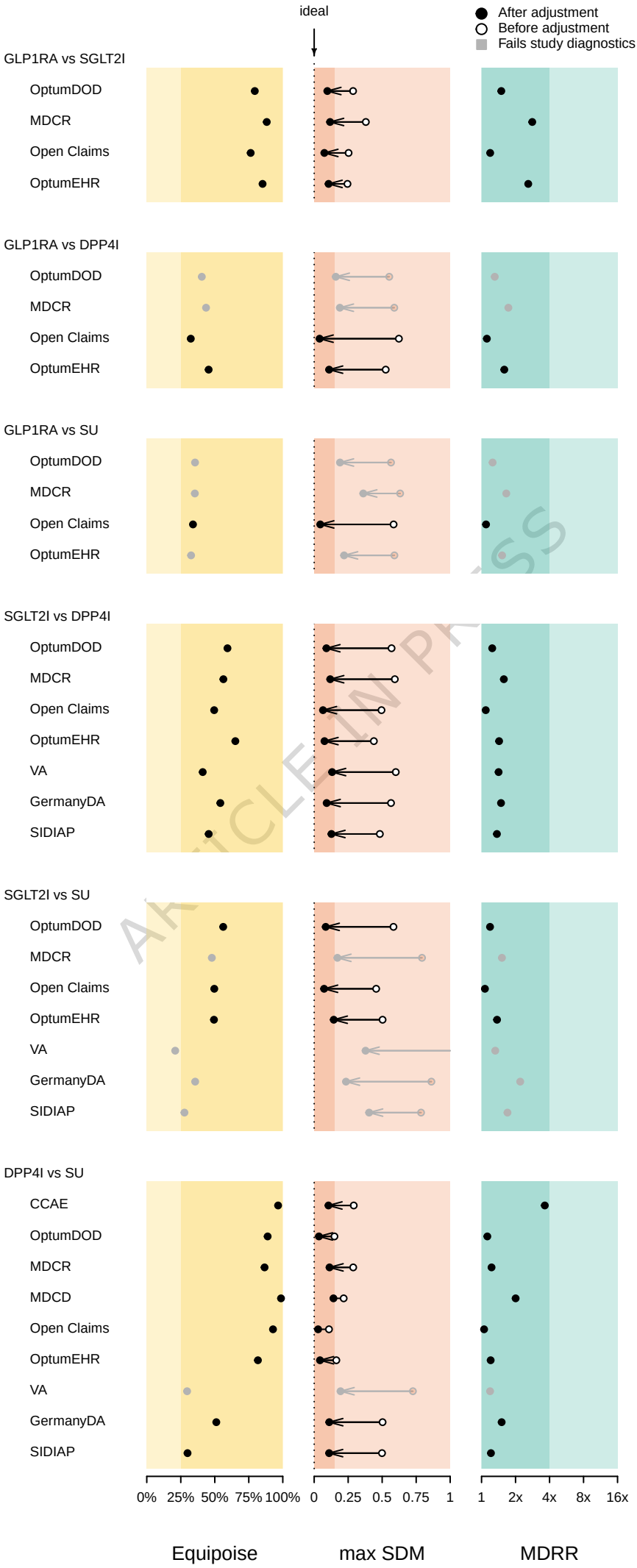
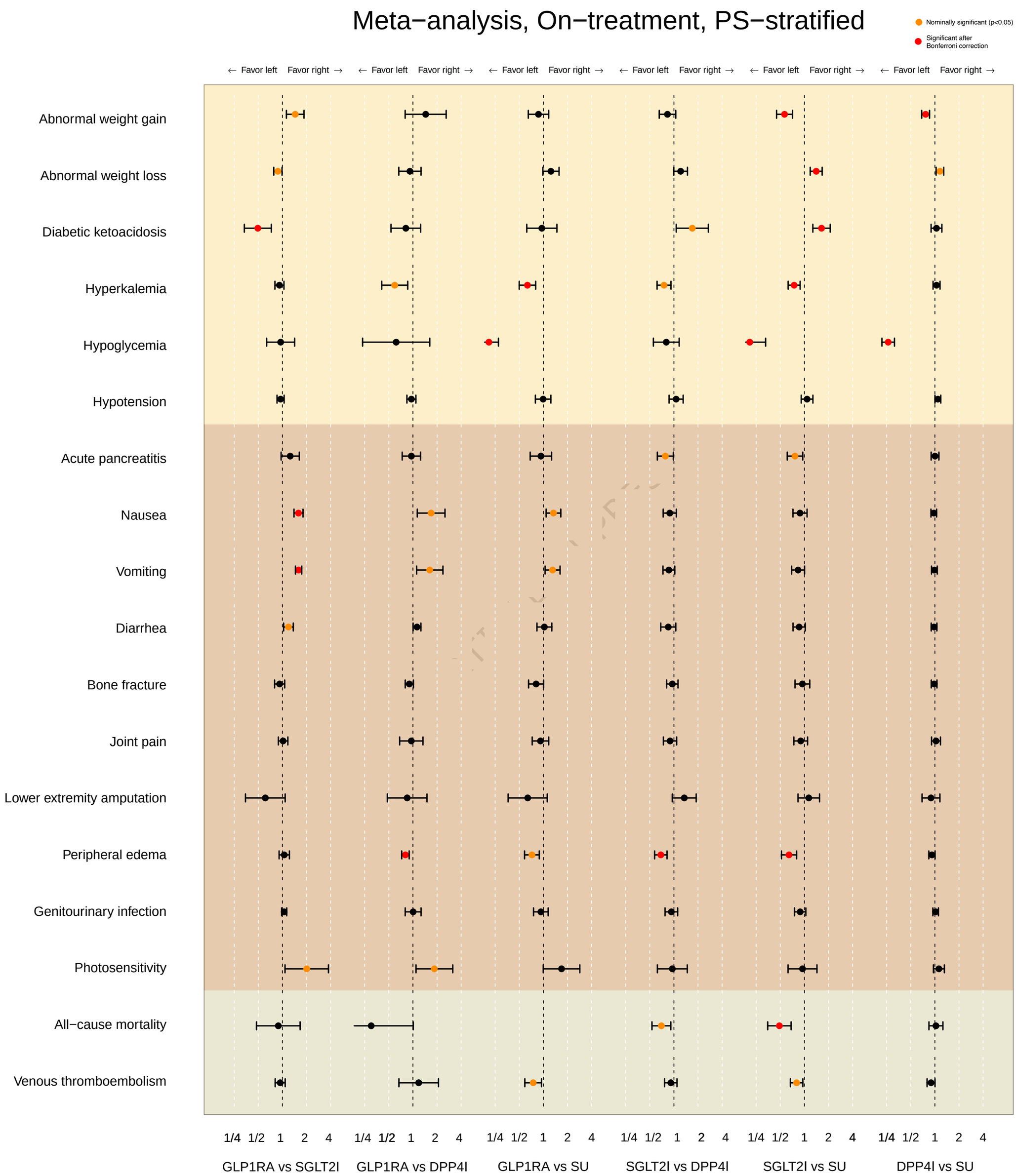
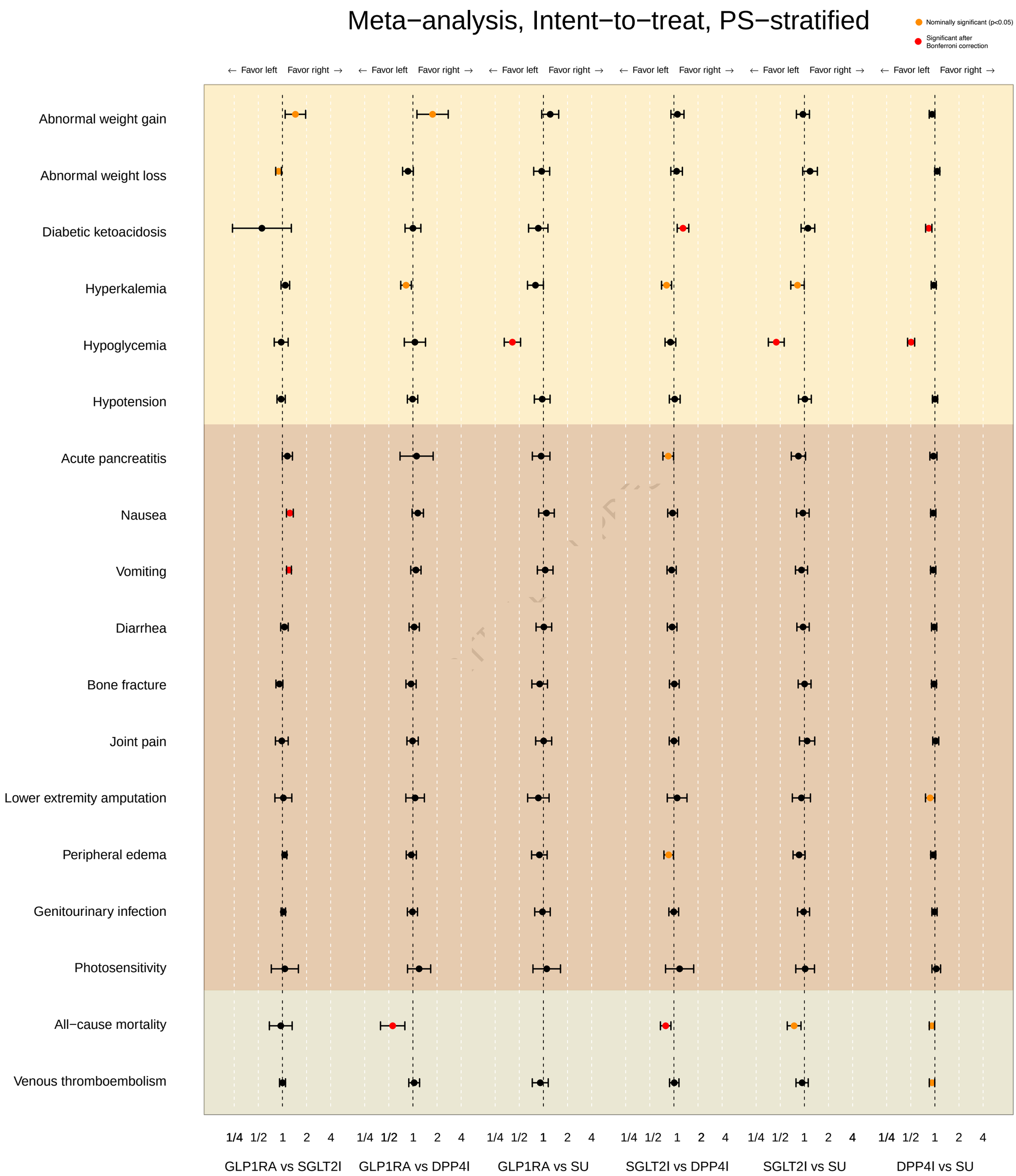



Meta−analysis, On−treatment, PS−stratified
Nominally significant (p<0.05)
Significant after Bonferroni correction
← Favor left Favor right →
Abnormal weight gain
Abnormal weight loss
Diabetic ketoacidosis
Hyperkalemia
Hypoglycemia
Hypotension
Acute pancreatitis
Nausea
Vomiting
Diarrhea
Bone fracture
Joint pain
Lower extremity amputation
Peripheral edema
Genitourinary infection
Photosensitivity
All−cause mortality
Venous thromboembolism
1/4 1/2 1 2 4
GLP1RA vs SGLT2I
GLP1RA vs DPP4I
GLP1RA vs SU
SGLT2I vs DPP4I
SGLT2I vs SU
DPP4I vs SU


Meta−analysis, Intent−to−treat, PS−stratified
Nominally significant (p<0.05)
Significant after Bonferroni correction
← Favor left Favor right →
Abnormal weight gain
Abnormal weight loss
Diabetic ketoacidosis
Hyperkalemia
Hypoglycemia
Hypotension
Acute pancreatitis
Nausea
Vomiting
Diarrhea
Bone fracture
Joint pain
Lower extremity amputation
Peripheral edema
Genitourinary infection
Photosensitivity
All−cause mortality
Venous thromboembolism
1/4 1/2 1 2 4
GLP1RA vs SGLT2I
GLP1RA vs DPP4I
GLP1RA vs SU
SGLT2I vs DPP4I
SGLT2I vs SU
DPP4I vs SU